

LAB#11**STRINGS AND STRING MANIPULATION IN C++****Lab Tasks**

1. Write a program to find a substring within a string. If found display its starting position.
2. Write program takes a string object from the user and calculates the number of vowels, digits and white-spaces.
3. Write a function to check whether a string “s” is a palindrome or not.
4. Write a program to convert a string in lowercase.
5. Write a C++ program to reverse a given string.
6. Write a C++ program to capitalize the first letter of each word of a given string. Words must be separated by only one space.
7. Write a C++ program to count all the words in a given string.

1. Write a program to find a substring within a string. If found display its starting position.

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    string string, sub;
    cout << "Enter string: ";
    getline(std::cin, string);
    cout << "Enter substring: ";
    getline(std::cin, sub);
    cout << endl;

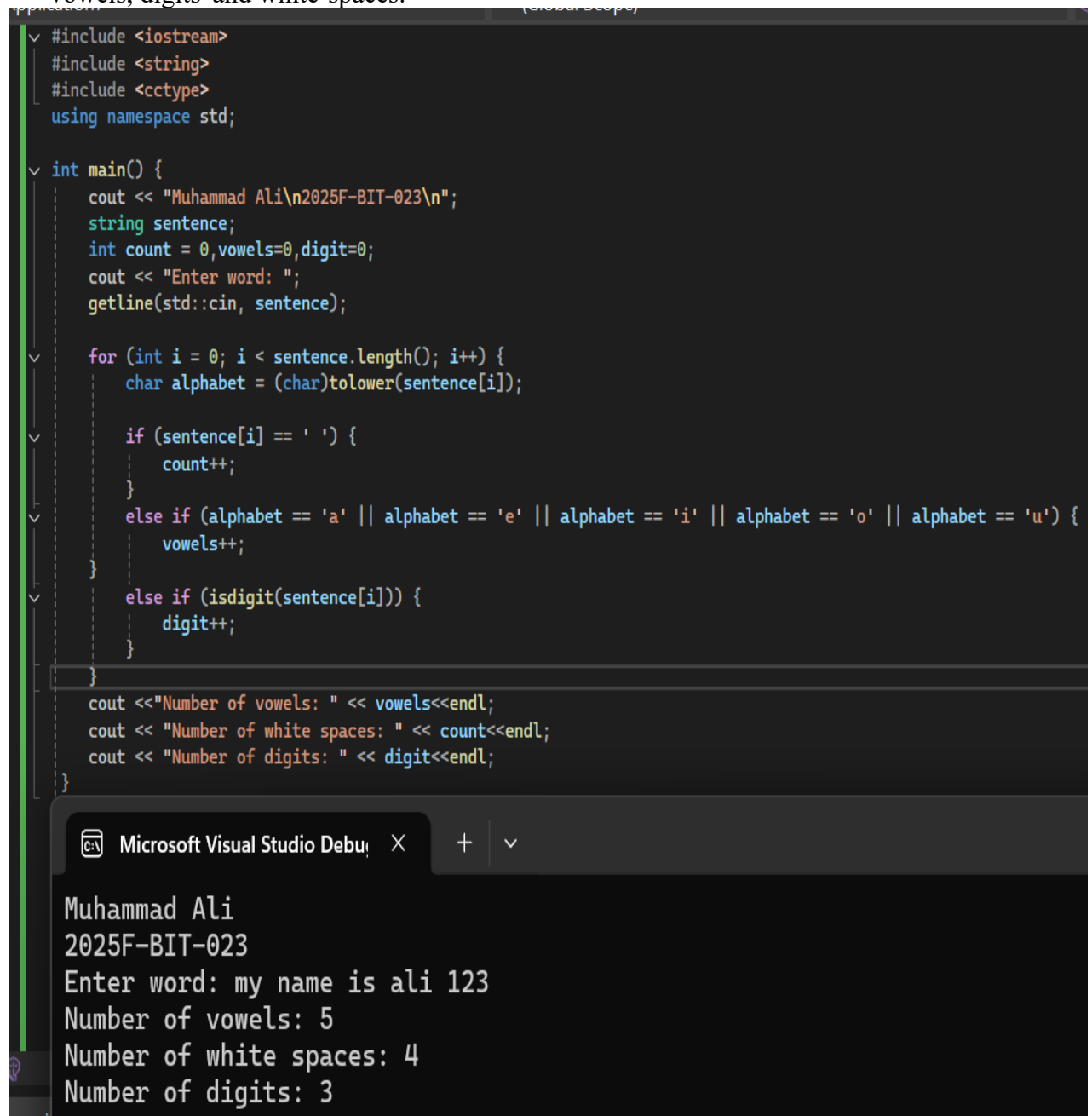
    int position = string.find(sub);

    if (position != string::npos) {
        cout << "Substring found at: " << position;
    }
    else {
        cout << "Substring not found";
    }
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter string: my name is ali
Enter substring: ali
Substring found at: 11

2. Write program takes a string object from the user and calculates the number of vowels, digits and white-spaces.



```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    string sentence;
    int count = 0, vowels=0, digit=0;
    cout << "Enter word: ";
    getline(std::cin, sentence);

    for (int i = 0; i < sentence.length(); i++) {
        char alphabet = (char)tolower(sentence[i]);

        if (sentence[i] == ' ') {
            count++;
        }
        else if (alphabet == 'a' || alphabet == 'e' || alphabet == 'i' || alphabet == 'o' || alphabet == 'u') {
            vowels++;
        }
        else if (isdigit(sentence[i])) {
            digit++;
        }
    }

    cout << "Number of vowels: " << vowels << endl;
    cout << "Number of white spaces: " << count << endl;
    cout << "Number of digits: " << digit << endl;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter word: my name is ali 123
Number of vowels: 5
Number of white spaces: 4
Number of digits: 3

3. Write a function to check whether a string "s" is a palindrome or not.

```
✓ #include <iostream>
  | #include <string>
  | #include <cctype>
  | using namespace std;
  |
  | void palindrome(string word) {
  |     string word2;
  |     for (int i = word.length() - 1; i >= 0; i--) {
  |         word2 += (word[i]);
  |     }
  |     cout << "Reversed word: " << word2 << endl;
  |
  |     if (word2 == word) {
  |         cout << "The word is palindrome";
  |     }
  |     else {
  |         cout << "The word is not palindrome";
  |     }
  | }
  |
  | int main() {
  |     cout << "Muhammad Ali\n2025F-BIT-023\n";
  |     string word;
  |     cout << "Enter word to check if palindrome or not: ";
  |     getline(std::cin, word);
  |     palindrome(word);
  |     cout << endl;
  | }
```

```
Muhammad Ali
2025F-BIT-023
Enter word to check if palindrome or not: madam
Reversed word: madam
The word is palindrome
```

4. Write a program to convert a string in lowercase.

```
✓ #include <iostream>
  #include <string>
  #include <cctype>
  using namespace std;

✓ int main() {
  cout << "Muhammad Ali\n2025F-BIT-023\n";
  string sentence;
  cout << "Enter Sentence: ";
  getline(std::cin, sentence);
  cout << endl << "Lowercase Sentence: ";
  for (int i = 0; i <= sentence.length(); i++) {
    cout << (char)tolower(sentence[i]);
  }
  cout << endl;
}
```



Microsoft Visual Studio Debug Console



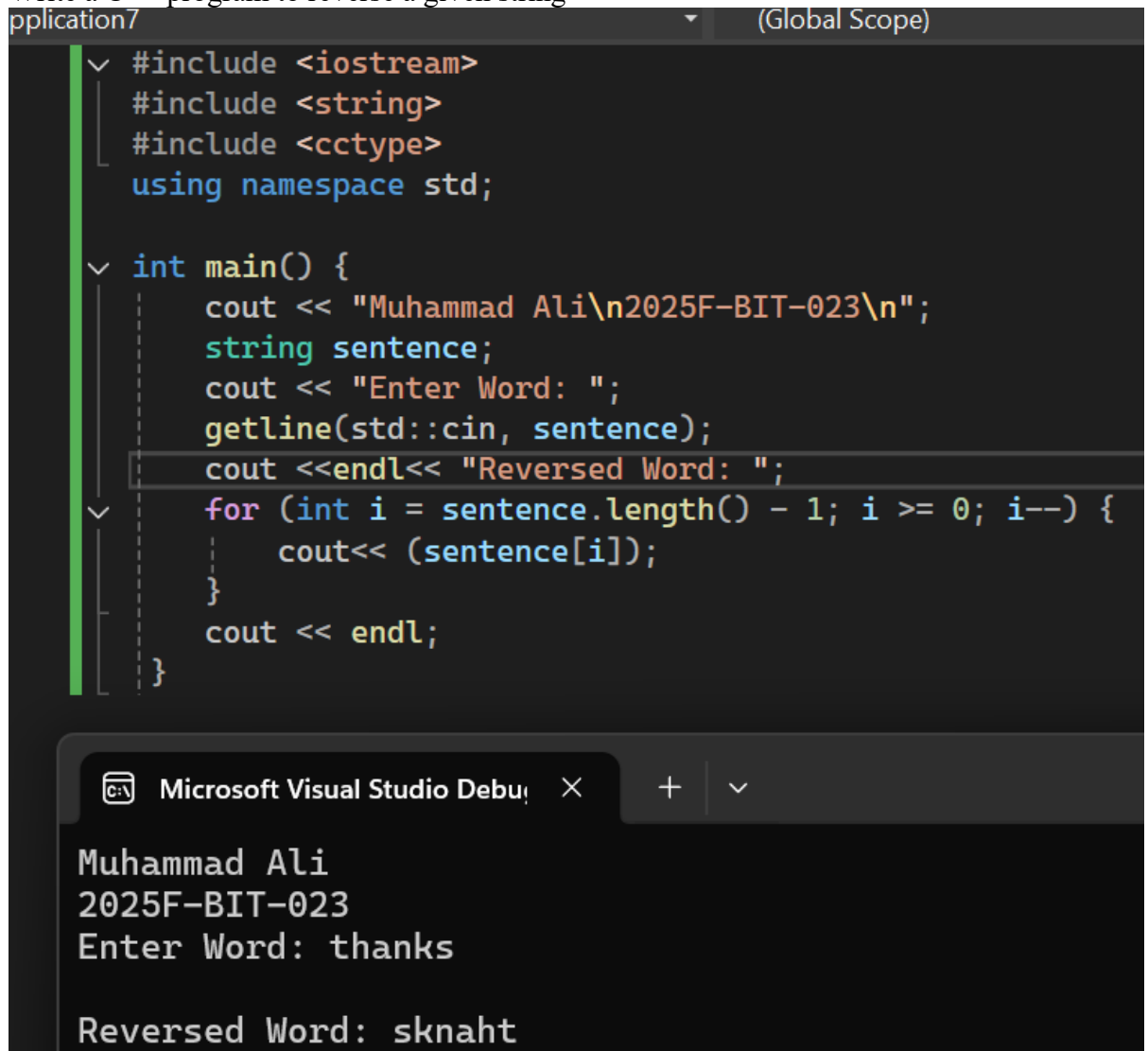
Muhammad Ali

2025F-BIT-023

Enter Sentence: MY NAME IS ALI

Lowercase Sentence: my name is ali

5. Write a C++ program to reverse a given string



The screenshot shows a C++ program in Visual Studio. The code is as follows:

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    string sentence;
    cout << "Enter Word: ";
    getline(std::cin, sentence);
    cout << endl << "Reversed Word: ";
    for (int i = sentence.length() - 1; i >= 0; i--) {
        cout << (sentence[i]);
    }
    cout << endl;
}
```

The output of the program is shown in the console window:

```
Muhammad Ali
2025F-BIT-023
Enter Word: thanks

Reversed Word: sknaht
```

6. Write a C++ program to capitalize the first letter of each word of a given string. Words must be separated by only one space.

```
1  #include <iostream>
2  #include <string>
3  #include <cctype>
4  using namespace std;
5
6  int main() {
7      cout << "Muhammad Ali\n2025F-BIT-023\n";
8      string sentence;
9      cout << "Enter sentence: ";
10     getline(std::cin, sentence);
11     sentence[0] = toupper(sentence[0]);
12     for (int i = 1; i < sentence.length(); i++) {
13         if (sentence[i - 1] == ' ') {
14             sentence[i] = toupper(sentence[i]);
15         }
16     }
17     cout << sentence<<endl;
18 }
19
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter sentence: my name is ali
My Name Is Ali

7. Write a C++ program to count all the words in a given string.

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    string sentence;
    int count = 0;
    cout << "Enter sentence: ";
    getline(std::cin, sentence);
    if (sentence.length() > 0) {
        count = 1;
        for (int i = 0; i < sentence.length(); i++) {
            if (sentence[i] == ' ') {
                count++;
            }
        }
    }
    cout << "The number of word in the sentence: " << count;
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter sentence: my name is ali
The number of word in the sentence: 4